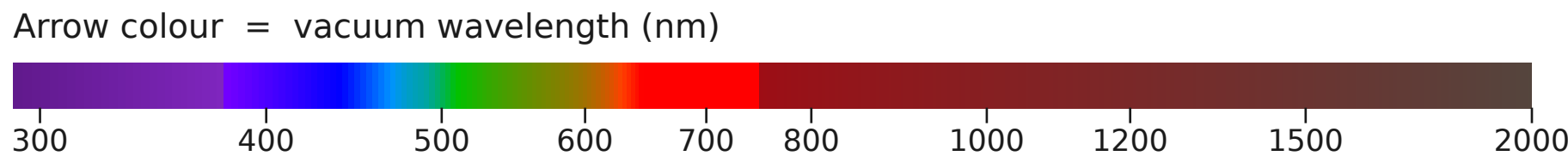


Antimony

Sb (⁵Sb I). Strongest classified transitions below 2 μm (NIST): 41 levels, 70 lines.

⁵Sb⁺ ionisation limit 69431.34 cm⁻¹ = 8.60839 eV (144.027 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J || e \mathbf{r} || J' \rangle$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J e \mathbf{r} J' \rangle $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
⁵ p ³ ⁴ S _{3/2} - ⁵ p ² 6s ⁴ p _{5/2}	206.9004	206.8344	1448.9696	2.179	1.810e+08	-	-	
⁵ p ³ ² D _{3/2} - ⁵ p ² 6s ² D _{3/2}	214.0373	213.9698	1400.6555	1.835	1.740e+08	-	-	
⁵ p ³ ⁴ S _{3/2} - ⁵ p ² 6s ⁴ p _{3/2}	217.6499	217.5818	1377.4066	1.887	1.750e+08	-	-	
⁵ p ³ ² D _{3/2} - ⁵ p ² 6s ² D _{5/2}	217.9872	217.9190	1375.2753	2.708	2.390e+08	-	-	
⁵ p ³ ⁴ S _{3/2} - ⁵ p ² 6s ⁴ p _{1/2}	231.2174	231.1463	1296.5825	1.436	1.690e+08	-	-	
⁵ p ³ ² D _{5/2} - ⁵ p ² 6s ² p _{3/2}	252.9269	252.8509	1185.2929	2.378	1.770e+08	-	-	
⁵ p ³ ² D _{3/2} - ⁵ p ² 6s ² p _{1/2}	259.8825	259.8048	1153.5694	0.603	2.100e+07	-	-	
⁵ p ³ ² D _{3/2} - ⁵ p ² 6s ⁴ p _{1/2}	287.8757	287.7913	1041.3954	0.999	4.240e+07	-	-	
⁵ p ³ ² p _{3/2} - ⁵ p ² 6s ² p _{3/2}	323.3428	323.2495	927.1661	1.431	3.070e+07	-	-	
⁵ p ³ ² p _{1/2} - ⁵ p ² 6s ² p _{1/2}	326.8433	326.7491	917.2360	1.306	4.950e+07	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).